



TRIOS CYBER

LABORATORY PRACTICE 23

File Integrity Monitoring Lab

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1. Laboratory Objective and Environment

The Day 23 activity validated Wazuh File Integrity Monitoring (FIM) on the Windows laboratory endpoint. A dedicated harmless directory under the user Documents folder was configured for real-time monitoring. A test file was then created and modified, and the resulting integrity-change event was verified in the Wazuh Dashboard.

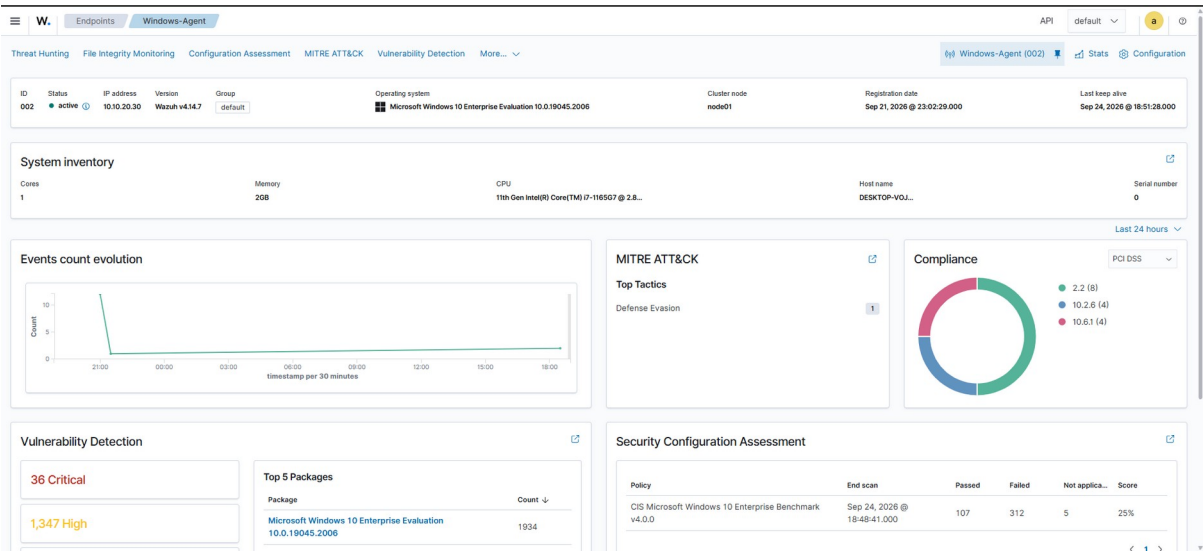
Field	Observed value	Interpretation
Wazuh server	trioscyber	Central monitoring platform
Endpoint	Windows-Agent / ID 002 / 10.10.20.30	Windows endpoint monitored by Wazuh
Monitored path	C:\Users\matthew\Documents\TriosFIMLab	Dedicated harmless laboratory directory
Test file	day23_test.txt	File used for controlled creation and modification

2. Wazuh Server and Windows Agent Readiness

Before the FIM test, the Wazuh server components were verified as active and the Windows WazuhSvc service was confirmed as Running. The Windows-Agent endpoint was also visible as Active in Wazuh, establishing a functional collection path before file-integrity testing.

```
matthew@trioscyber:~$ sudo systemctl restart wazuh-indexer
sudo systemctl restart wazuh-manager
sudo systemctl restart filebeat
sudo systemctl restart wazuh-dashboard
[sudo] password for matthew:
matthew@trioscyber:~$ echo "===== WAZUH SERVICES ====="
systemctl is-active wazuh-indexer
systemctl is-active wazuh-manager
systemctl is-active filebeat
systemctl is-active wazuh-dashboard
===== WAZUH SERVICES =====
active
active
active
active
```

```
matthew@DESKTOP-VOJCD50 C:\Users\matthew>powershell.exe -NoProfile -Command "Restart-Service WazuhSvc; Start-Sleep -Seconds 3; Get-Service WazuhSvc"
Status Name DisplayName
-----
Running WazuhSvc Wazuh
```



3. Safe Test Directory and Configuration Backup

A dedicated folder, C:\Users\matthew\Documents\TriosFIMLab, was created specifically for the laboratory. Before changing the Wazuh agent configuration, ossec.conf was copied to ossec.conf.day23.bak so the original configuration could be restored if required.

```
matthew@DESKTOP-VOJCD50 C:\Users\matthew>mkdir C:\Users\matthew\Documents\TriosFIMLab
matthew@DESKTOP-VOJCD50 C:\Users\matthew>copy "C:\Program Files (x86)\ossec-agent\ossec.conf" "C:\Program Files (x86)\ossec-agent\ossec.conf.day23.bak"
1 archivo(s) copiado(s).
```

4. Real-Time FIM Monitoring Configuration

The test directory was added to the Wazuh agent syscheck configuration with realtime="yes". A findstr verification confirmed that C:\Users\matthew\Documents\TriosFIMLab was present in ossec.conf as a real-time monitored directory.

```
matthew@DESKTOP-VOJCD50 C:\Users\matthew>powershell.exe -NoProfile -Command "$p='C:\Program Files (x86)\ossec-agent\ossec.conf'; $c=Get-Content $p -Raw; if($c -notmatch [regex]::Escape('C:\Users\matthew\Documents\TriosFIMLab')){$c=$c.Replace('</syscheck>', ' <directories realtime='yes'>C:\Users\matthew\Documents\TriosFIMLab</directories>' + [Environment]::NewLine + '</syscheck>'); Set-Content $p $c -Encoding UTF8}"
matthew@DESKTOP-VOJCD50 C:\Users\matthew>findstr /i "TriosFIMLab" "C:\Program Files (x86)\ossec-agent\ossec.conf"
<directories realtime="yes">C:\Users\matthew\Documents\TriosFIMLab</directories>
```

5. Windows Agent Restart and Configuration Reload

The Windows Wazuh service was stopped and started with `sc.exe` so the modified FIM configuration could be loaded. The service transitioned through STOPPED and START_PENDING states and the final query confirmed state 4 (RUNNING).

```
matthew@DESKTOP-VOJCDSO C:\Users\matthew>sc.exe queryex WazuhSvc

NOMBRE_DE_SERVICIO: WazuhSvc
TIPO                : 10  WIN32_OWN_PROCESS
ESTADO              : 1  STOPPED
CÓDIGO_DE_SALIDA_DE_WIN32 : 0  (0x0)
CÓDIGO_DE_SALIDA_DEL_SERVICIO: 0  (0x0)
PUNTO_DE_CONTROL    : 0x0
ESPERA              : 0x0
PID                 : 0
MARCAS              :

matthew@DESKTOP-VOJCDSO C:\Users\matthew>sc.exe start WazuhSvc

NOMBRE_SERVICIO: WazuhSvc
TIPO            : 10  WIN32_OWN_PROCESS
ESTADO          : 2  START_PENDING
                 (NOT_STOPPABLE, NOT_PAUSABLE, IGNORES_SHUTDOWN)
CÓD_SALIDA_WIN32 : 0  (0x0)
CÓD_SALIDA_SERVICIO: 0  (0x0)
PUNTO_COMPROB.  : 0x0
INDICACIÓN_INICIO : 0x7d0
PID             : 13276
MARCAS          :

matthew@DESKTOP-VOJCDSO C:\Users\matthew>timeout /t 5 /nobreak >nul

matthew@DESKTOP-VOJCDSO C:\Users\matthew>sc.exe query WazuhSvc

NOMBRE_SERVICIO: WazuhSvc

matthew@DESKTOP-VOJCDSO C:\Users\matthew>sc.exe query WazuhSvc

NOMBRE_SERVICIO: WazuhSvc
TIPO            : 10  WIN32_OWN_PROCESS
ESTADO          : 4  RUNNING
                 (STOPPABLE, NOT_PAUSABLE, IGNORES_SHUTDOWN)
CÓD_SALIDA_WIN32 : 0  (0x0)
CÓD_SALIDA_SERVICIO: 0  (0x0)
PUNTO_COMPROB.  : 0x0
INDICACIÓN_INICIO : 0x0
```

6. Harmless Test File Creation and Modification

A harmless text file named `day23_test.txt` was created inside the monitored TriosFIMLab directory with initial content. A second line was then appended to the same file, producing a controlled content modification without altering any critical Windows file.

```
matthew@DESKTOP-VOJCDSO C:\Users\matthew>echo TRIOS CYBER DAY 23 - Initial content > C:\Users\matthew\Documents\TriosFIMLab\day23_test.txt

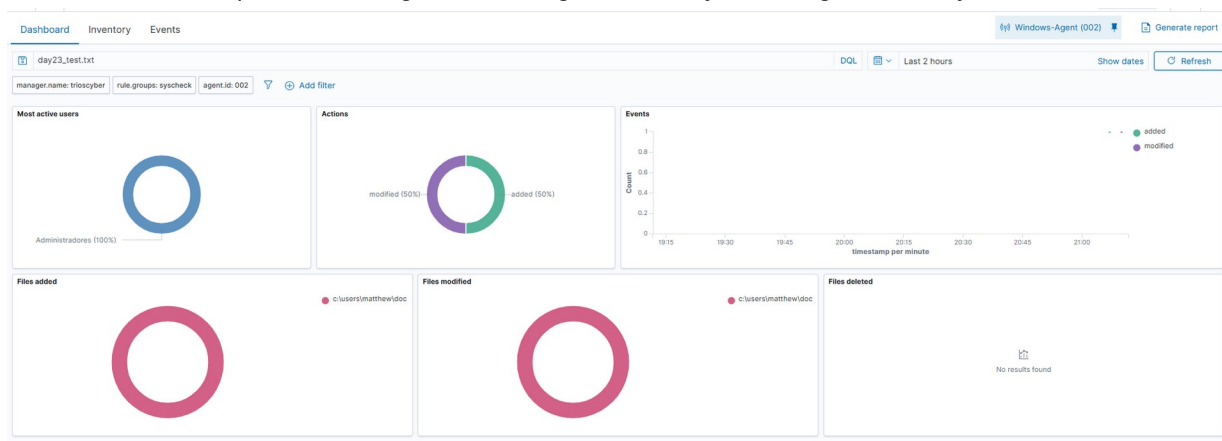
matthew@DESKTOP-VOJCDSO C:\Users\matthew>type C:\Users\matthew\Documents\TriosFIMLab\day23_test.txt
TRIOS CYBER DAY 23 - Initial content

matthew@DESKTOP-VOJCDSO C:\Users\matthew>echo Modified content detected by Wazuh FIM >> C:\Users\matthew\Documents\TriosFIMLab\day23_test.txt

matthew@DESKTOP-VOJCDSO C:\Users\matthew>type C:\Users\matthew\Documents\TriosFIMLab\day23_test.txt
TRIOS CYBER DAY 23 - Initial content
Modified content detected by Wazuh FIM
```

7. File Integrity Monitoring Detection

The File Integrity Monitoring view for Windows-Agent was refreshed after the test activity. The dashboard registered file-integrity activity for the monitored endpoint, confirming that the configured directory was being observed by Wazuh FIM.



8. FIM Event Details and Integrity Change Verification

The resulting Wazuh document identified Windows-Agent (ID 002) at 10.10.20.30 as the source endpoint and showed the monitored file C:\Users\matthew\Documents\TriosFIMLab\day23_test.txt as modified. The event used the syscheck_integrity_changed decoder, syscheck mode realtime, rule ID 550 at level 7, and the description Integrity checksum changed. Wazuh also mapped the alert to MITRE ATT&CK T1565.001 (Stored Data Manipulation) under the Impact tactic.

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TableJSON

index	wazuh-alerts-4.x-2026.09.25
agent.id	002
agent.ip	10.10.20.30
agent.name	Windows-Agent
decoder.name	syscheck_integrity_changed
full_log	File 'c:\users\matthew\documents\triosfimlab\day23_test.txt' modified Mode: realtime Changed attributes: size, mtime, md5, sha1, sha256 Size changed from '80' to '113' Old modification time was: '1790302054', now it is '1790302233' Old md5sum was: '9fa301d3c1535d9926aaf7189316f02' New md5sum is: '4865668a965a5a0913a345cca5eacc40' New sha1 is: '204d3d810ff670646158352bcaeb2f87d7b2f3' New sha256 is: 'ec91see7f68e475480cc50b78f9d1445d1bc45734c2857db35a80751d04968c'
id	1790302236.316483
input.type	log
location	syscheck
manager.name	trioscyber
rule.description	Integrity checksum changed.
rule.firedtimes	1
rule.gdpr	II.5.1.f
rule.ppg13	4.11
rule.groups	ossec, syscheck, syscheck_entry_modified, syscheck_file
rule.hipaa	164.312.c.1, 164.312.c.2
rule.id	550
rule.level	7
rule.mal	false
rule.mitre.id	T1565.001
rule.mitre.tactic	Impact

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rule.mitre.tactic	Impact
rule.mitre.technique	Stored Data Manipulation
rule.nist_800_53	SI.7
rule.pci_dss	11.5
rule.tsc	PII.4, PII.5, CO6.1, CO6.8, CC7.2, CC7.3
syscheck.attrs_after	ARCHIVE
syscheck.changed_attributes	size, mtime, md5, sha1, sha256
syscheck.event	modified
syscheck.md5_after	4865668a965a5a0913a345cca5eacc40
syscheck.md5_before	9fa301d3c1535d9926aaf7189316f02
syscheck.mode	realtime
syscheck.mtime_after	Sep 24, 2026 @ 21:10:33.000
syscheck.mtime_before	Sep 24, 2026 @ 21:07:34.000
syscheck.path	c:\users\matthew\documents\triosfimlab\day23_test.txt
syscheck.sha1_after	204d3d810ff670646158352bcaeb2f87d7b2f3
syscheck.sha1_before	fdbcbfc305c74a7c94ef757a0b39bb200338c1d3
syscheck.sha256_after	ec91see7f68e475480cc50b78f9d1445d1bc45734c2857db35a80751d04968c
syscheck.sha256_before	b99ce48a98dd26181b55f592b87f31be3c029181f2732581c791aba8a914d1d7
syscheck.size_after	113
syscheck.size_before	80
syscheck.uid_after	S-1-5-32-544
syscheck.uname_after	Administradores
syscheck.win_perm_after.allo	DELETE, READ_CONTROL, WRITE_DAC, WRITE_OWNER, SYNCHRONIZE, READ_DATA, WRITE_DATA, APPEND_DATA, READ_EA, WRITE_EA, EXECUTE, READ_ATTRIBUTES, WRITE_ATTRIBUTES, DELETE, READ_CONTROL, WRITE_DAC, WRITE_OWN
syscheck.win_perm_after.name	SYSTEM, Administradores, matthew
timestamp	Sep 24, 2026 @ 21:10:36.667

9. FIM Validation Summary

Field	Observed value	Interpretation
Source endpoint	Windows-Agent / ID 002 / 10.10.20.30	Monitored Windows endpoint identified
Path	C:\Users\matthew\Documents\TriosFIMLab\day23_test.txt	Dedicated non-critical test file
FIM event	modified / realtime	File content change detected in real time
Wazuh rule	550 / level 7	Integrity checksum changed
MITRE ATT&CK	T1565.001 - Stored Data Manipulation	Impact mapping associated with the detection

10. Conclusions

The File Integrity Monitoring lab was completed successfully. A dedicated harmless directory was configured for real-time monitoring, day23_test.txt was created and modified, and Wazuh detected the integrity change from Windows-Agent. The event preserved the monitored path, modification state, integrity metadata, rule information, and MITRE ATT&CK mapping, satisfying the Day 23 requirement without modifying critical system files.